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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2115363240

CHEMISTRY **5070/43**

Paper 4 Alternative to Practical **October/November 2010**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

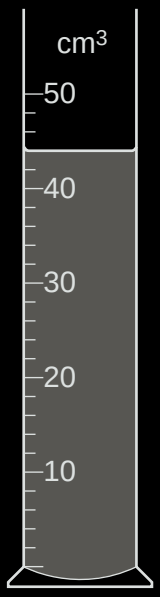
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use

This document consists of **15** printed pages and **1** blank page.

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(a) **(i)** Name the apparatus.

..... [1]

(ii) What is the volume of 0.100 mol/dm^3 sulfuric acid?

..... cm^3 [1]

(iii) Using your answer to **(a)(ii)**, calculate the number of moles of 0.100 mol/dm^3 sulfuric acid.

..... moles [1]

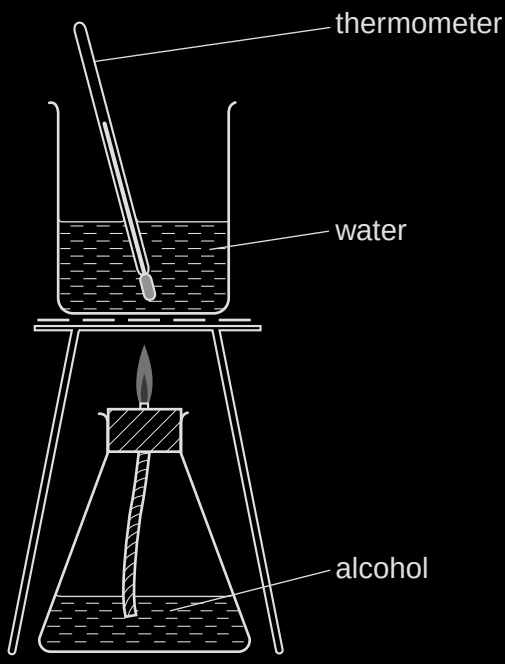
(b) **(i)** The sulfuric acid was poured into a beaker and 0.12 g of magnesium was added. The magnesium reacted with the sulfuric acid and hydrogen was produced.

How many moles of magnesium were added?
[A_r: Mg, 24]

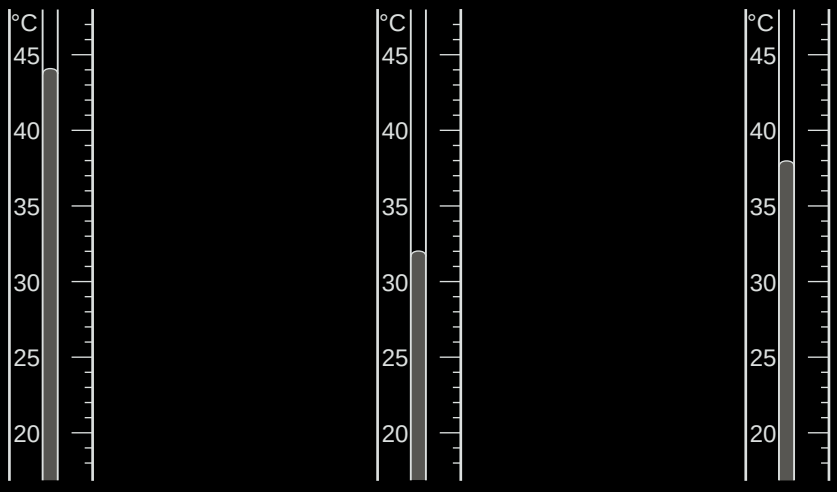
..... moles [1]

(ii) Write the equation for the reaction between magnesium and sulfuric acid.

..... [1]



(a) (i) The initial temperature of the water was, in each case, 20 °C.
The diagrams below show parts of the thermometer stem giving the temperature of the water after the burning of each alcohol.

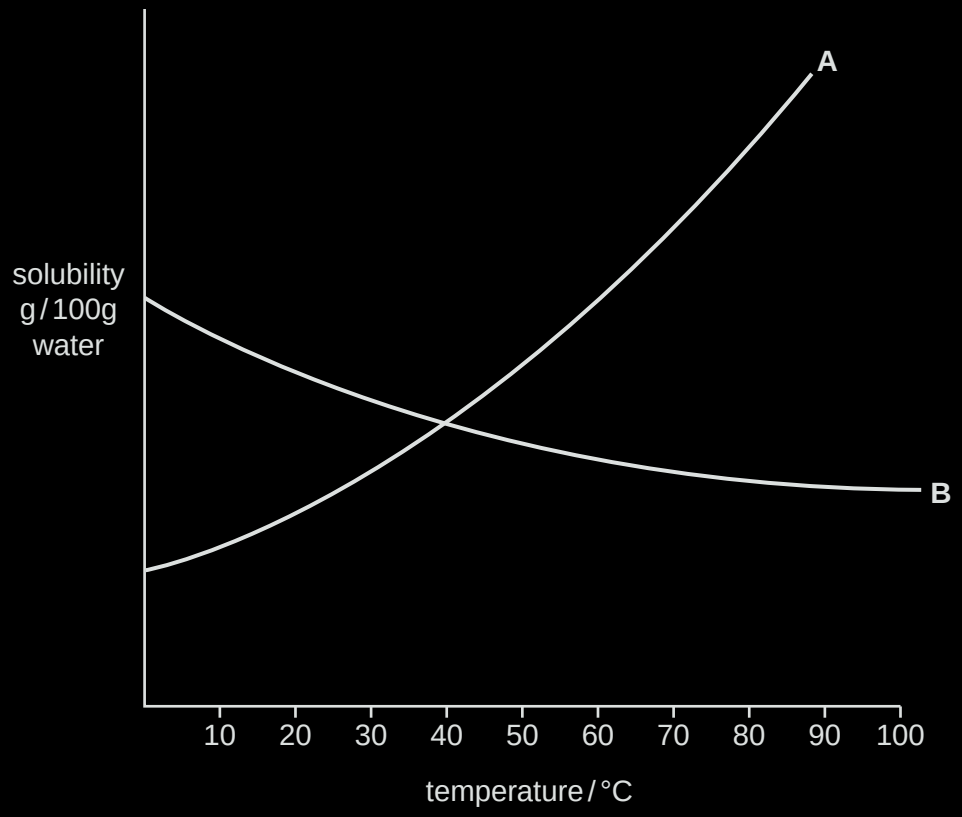


alcohol **X** alcohol **Y** alcohol **Z**

Use the thermometer readings to complete the following table and calculate the rise in temperature for each alcohol.

	alcohol **X**	alcohol **Y**	alcohol **Z**
final temperature / °C			
initial temperature / °C	20	20	20
rise in temperature / °C			

[3]



Which one of the following conclusions is correct?

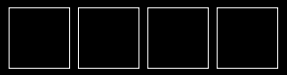
(a) **A** is more soluble than **B** at all temperatures.

(b) **B** is more soluble than **A** below 40 °C.

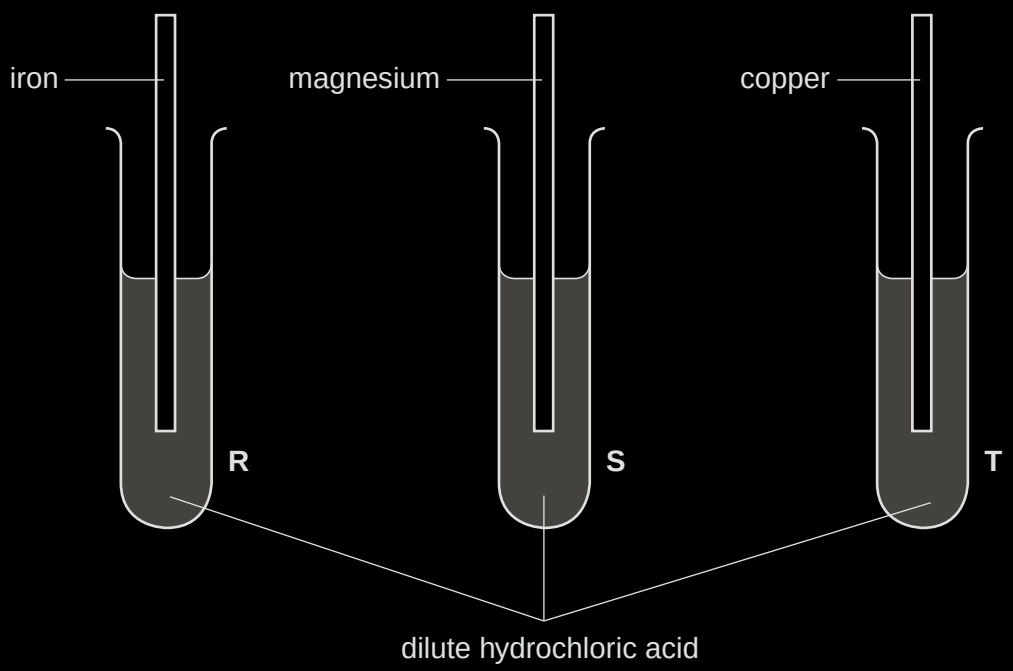
(c) **A** is less soluble than **B** above 40 °C.

(d) The solubility of both salts increases with increasing temperature. [1]

[Total: 1]







In which test-tubes was hydrogen gas produced?

(a) **R** and **S** only ☐

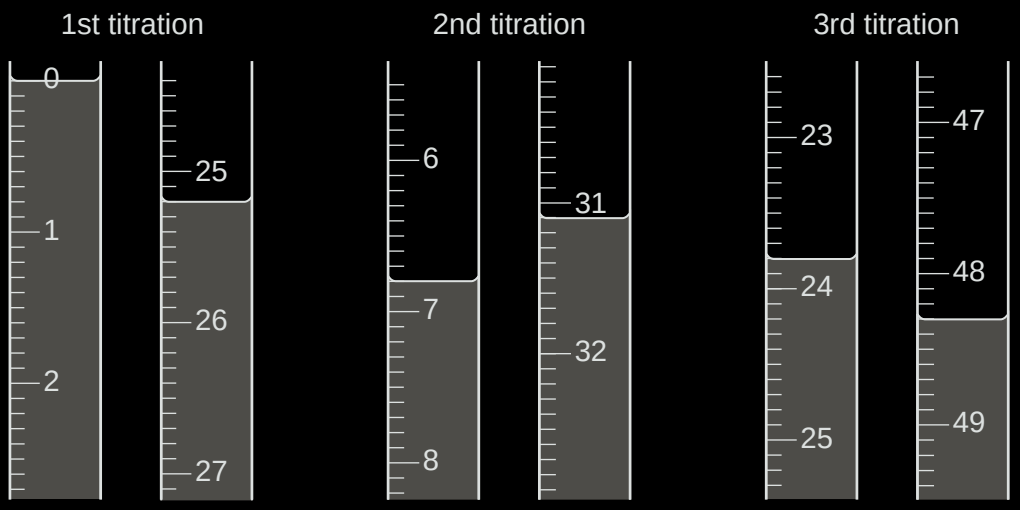
(b) **R** and **T** only ☐

(c) **S** and **T** only ☐

(d) **R** and **S** and **T** ☐

[1]

[Total: 1]



(c) Use the diagrams to complete the following table.

titration number	1	2	3
final burette reading / cm^3			
initial burette reading / cm^3			
volume of hydrochloric acid / cm^3			
best titration results (✓)			

Summary

Tick (✓) the best titration results.
Using these results, the average volume of hydrochloric acid was

..... cm^3. [4]

(d) Calculate the number of moles of hydrochloric acid in the average volume of 0.100 mol/dm^3 hydrochloric acid calculated in **(c)**.

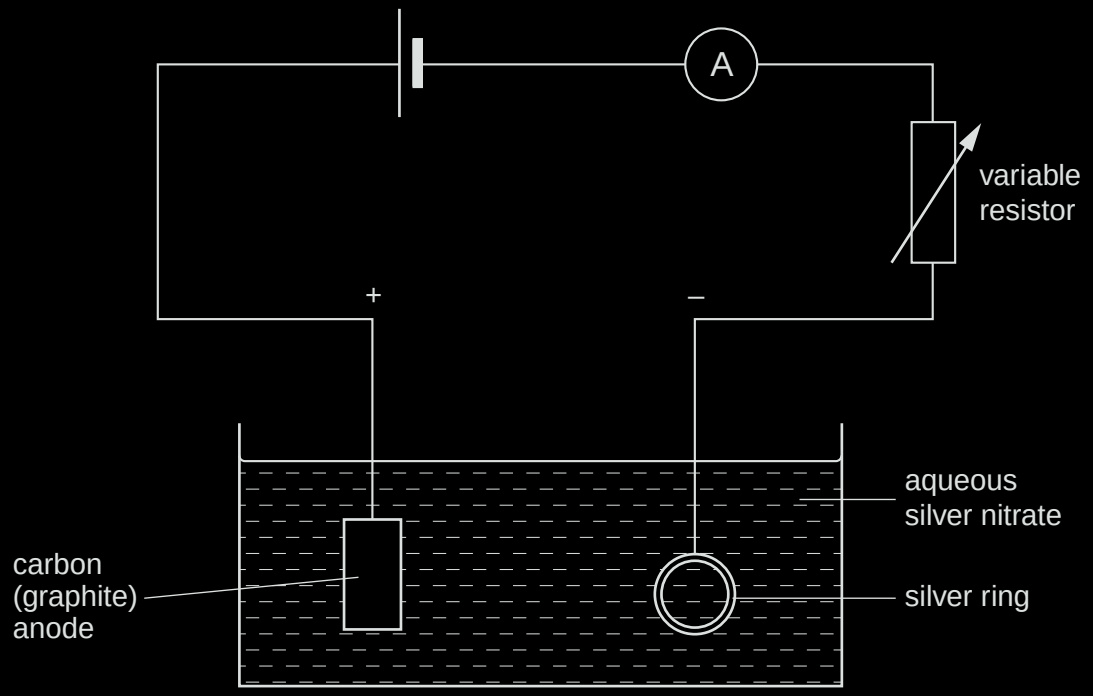
..... moles [1]

(e) Hydrochloric acid reacts with sodium hydroxide according to the following equation.

$$HCl + NaOH \rightarrow NaCl + H_2O$$

Deduce the number of moles of sodium hydroxide present in 25.0 cm^3 of **W**.

.....moles [1]



The ring, which was the cathode, was weighed before it was placed in the aqueous silver nitrate. The circuit was completed and a current of 1.0 A was allowed to flow.

The ring was removed every ten minutes, washed, dried and weighed before being returned to the solution and reconnected to the circuit.
This was experiment **1**.

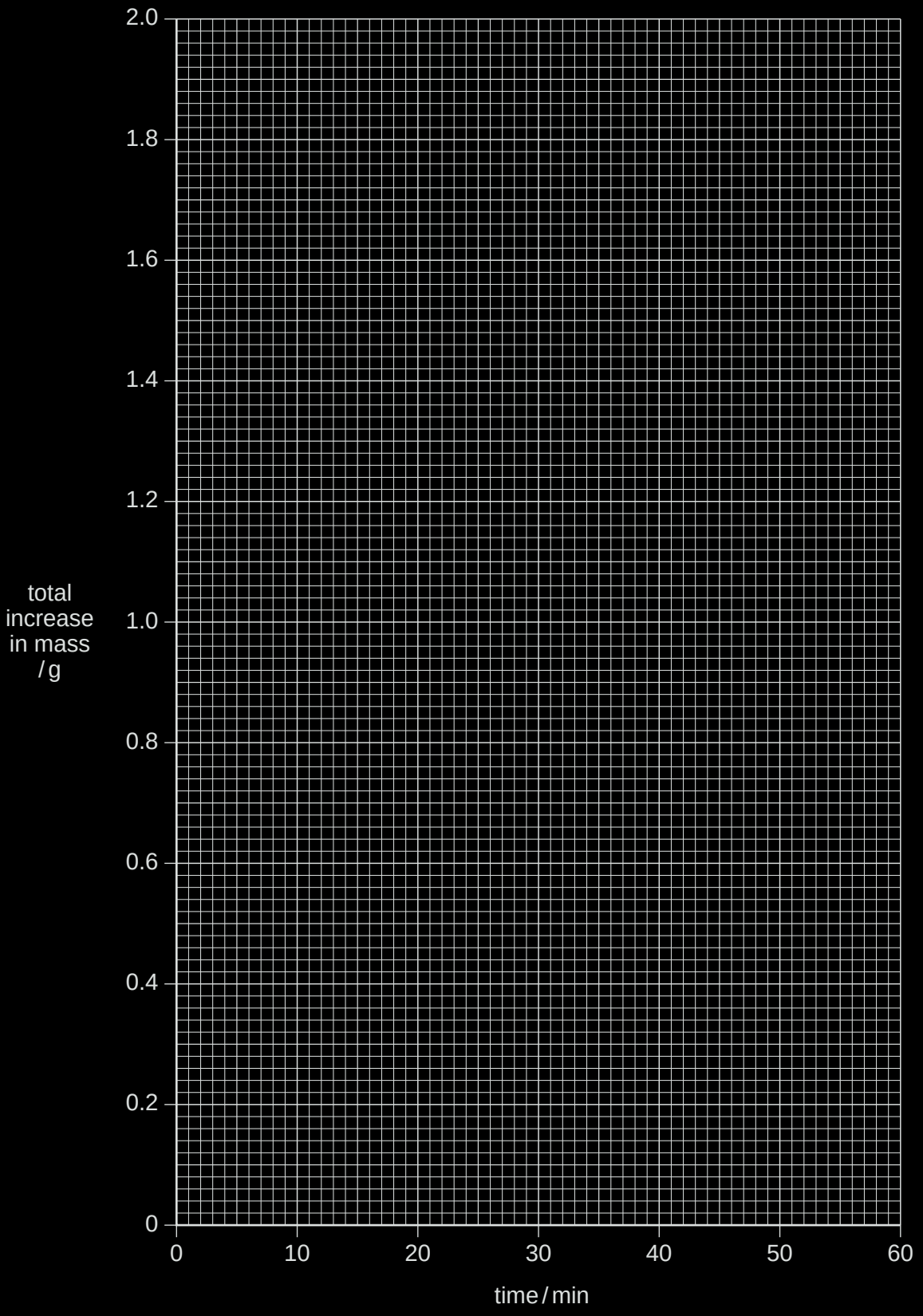
The experiment was repeated using a current of 1.5 A. This was experiment **2**.

The results from both experiments are shown in the table below.

(a) Complete the table by calculating the total increase in mass after each ten minute period in both experiments.

	experiment **1**		experiment **2**	
	current 1.0 A		current 1.5 A	
time / mins	mass of ring / g	total increase in mass / g	mass of ring / g	total increase in mass / g
0	8.80	0.00	8.80	0.00
10	9.10	0.30	9.20	0.40
20	9.40	0.60	9.60	
30	9.70		10.00	
40	10.00		10.40	
50	10.30		10.40	

[1]


2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
total increase in mass / g
0
10
20
30
40
50
60
time / min

[4]